**College of Engineering – Department of Mechanical Engineering**

# CURRICULUM

**Bachelor of Science in Mechanical Engineering**
Academic Year 2025-2026
Reference CMOs: CMO No. 97 s. 2017, CMO No. 20, s. 2013, CMO No. 39, s. 2021, and CMO 40, s. 2021

## Curriculum Description

Mechanical Engineering is a profession that concerns itself with mechanical design, energy conversion, fuel and combustion technologies, heat transfer, materials, noise control, and acoustics, manufacturing processes, rail transportation, automatic control, product safety and reliability, solar energy, and technological impacts to society. Mechanical engineers study the behavior of materials when forces are applied to them, such as the motion of solids, liquids, gasses, and heating and cooling of objects and machines. Using these basic building blocks, mechanical engineers design space vehicles, computers, power plants, intelligent machines and robots, automobiles, trains, airplanes, furnaces, and air conditioners. Mechanical engineers work on jet engine design, submarines, hot air balloons, textiles and new materials, medical and hospital equipment, and refrigerators and other home appliances. Anything that is mechanical or must interact with another machine or human being is within the broad scope of mechanical engineering.

The program also aims to develop graduates with ethical behavior, integrity, social responsibility, effective communication, teamwork, and leadership skills, while promoting lifelong learning to keep up with the continuing technological advancement.

Graduates of this program are expected to be engaged in design, manufacturing, thermal and mechanical systems, refrigeration and air conditioning, material handling, manufacturing, safety, engineering management, energy and environmental systems, automation, business, consultancy, academe, research and development, and other related industries.

## Program Education Objectives

The mechanical engineering alumni three to five years after graduation shall:
PEO 1: Practice as a specialist in solving complex mechanical engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, economic, and sustainability requirements.
PEO 2: Assume leadership position in multidisciplinary workforce in industry, academe, government, or private sector both local and global with consideration to social and ethical responsibilities.
PEO 3: Engage in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

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**Student Outcomes**

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:

**SO1 Discipline Knowledge.** Ability to apply mathematics, sciences and principles of engineering to solve complex mechanical engineering problems;

**SO2 Investigation.** Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;

**SO3 Design/Development of Solutions.** Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.

**SO4 Leadership and Teamwork.** Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

**SO5 Problem Analysis.** Identify, formulate, and solve complex mechanical engineering problems by applying principles of engineering, science, and mathematics;

**SO6 Ethics and Professionalism.** Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.

**SO7 Communication.** Communicate effectively on complex mechanical engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

**SO8 Environment and Sustainability.** Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

**SO9 Lifelong Learning.** Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

**SO10 The Engineer and Society.** Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

**SO11 Modern Tool Usage.** Apply appropriate techniques, skills, and modern engineering and IT tools to complex mechanical engineering activities;

**SO12 Project Management and Finance.** Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

**SO13 Social and National Responsibility.** Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

**CURRICULUM COMPONENTS**

| Classification / Field / Course | Number of Hours per Week | | Credit Units |
|---|---|---|---|
| | Lec | Lab | |
| **I. TECHNICAL COURSES** | | | |
| **A. Mathematics** | | | |

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| | | | |
|---|---|---|---|
| Differential Calculus | 4 | 0 | 4 |
| Integral Calculus | 4 | 0 | 4 |
| Probability and Statistics for AI | 3 | 0 | 3 |
| Differential Equations | 3 | 0 | 3 |
| Numerical Methods | 3 | 0 | 3 |
| **Subtotal** | **17** | **0** | **17** |
| **B. Natural/ Physical Sciences** | | | |
| Chemistry for Engineers | 3 | 3 | 4 |
| Physics for Engineers 1 | 3 | 3 | 4 |
| Physics for Engineers 2 | 2 | 3 | 3 |
| **Subtotal** | **8** | **9** | **11** |
| **C. Basic Engineering Sciences** | | | |
| Engineering Drawing | 0 | 3 | 1 |
| Computer-Aided Design | 0 | 3 | 1 |
| Computer Programming | 0 | 3 | 1 |
| Statics of Rigid Bodies | 3 | 0 | 3 |
| Dynamics of Rigid Bodies | 2 | 0 | 2 |
| Mechanics of Deformable Bodies | 4 | 0 | 4 |
| Engineering Economics | 3 | 0 | 3 |
| Engineering Management | 2 | 0 | 2 |
| Technopreneurship | 3 | 0 | 3 |
| Introduction to Engineering | 0 | 3 | 1 |
| Environmental Science and Engineering | 3 | 0 | 3 |
| **Subtotal** | **20** | **12** | **24** |
| **D. Allied Courses** | | | |
| Basic Electrical Engineering | 2 | 3 | 3 |
| Basic Electronics | 2 | 3 | 3 |
| DC and AC Machinery | 2 | 3 | 3 |
| **Subtotal** | **6** | **9** | **9** |
| **E. Fundamental ME Courses** | | | |
| Mechanical Engineering Orientation | 1 | 0 | 1 |
| Thermodynamics 1 | 3 | 0 | 3 |
| Workshop Theory and Practice | 0 | 3 | 1 |
| Advanced Mathematics for ME | 3 | 0 | 3 |
| Thermodynamics 2 | 3 | 0 | 3 |
| Fluid Mechanics for ME | 3 | 0 | 3 |
| Computer Applications for ME | 0 | 3 | 1 |
| Machine Shop Theory and Practice | 0 | 6 | 2 |
| Heat Transfer | 3 | 0 | 3 |
| Combustion Engineering | 3 | 0 | 3 |
| Fluid Machineries | 3 | 0 | 3 |
| Machine Elements | 2 | 3 | 3 |
| Materials Science and Engineering for ME | 2 | 3 | 3 |
| Mechanical Engineering Laboratory 1 | 0 | 3 | 1 |
| Refrigeration Systems | 3 | 0 | 3 |
| Vibration Engineering | 2 | 0 | 2 |
| Airconditioning and Ventilation Systems | 2 | 3 | 3 |
| ME Laboratory 2 | 0 | 6 | 2 |

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| | | | |
|---|---|---|---|
| Manufacturing and Industrial Processes with Plant Visits | 1 | 3 | 2 |
| ME Practice with Comprehensive Examination | 0 | 6 | 2 |
| Control Systems Engineering | 2 | 3 | 3 |
| Basic Occupational Safety and Health | 3 | 0 | 3 |
| Methods of Research | 3 | 0 | 3 |
| OJT/Internship (320 hrs) | | | 4 |
| **Subtotal** | **39** | **42** | **60** |
| **F. Professional ME Courses** | | | |
| Machine Design 1 | 3 | 0 | 3 |
| ME Laws, Ethics, Contracts, Codes and Standards | 2 | 0 | 2 |
| ME Project Study 1 | 0 | 3 | 1 |
| Machine Design 2 | 2 | 3 | 3 |
| Power Plant Design with Renewable Energy | 4 | 3 | 5 |
| Industrial Plant Engineering | 3 | 3 | 4 |
| ME Laboratory 3 | 0 | 6 | 2 |
| ME Project Study 2 | 0 | 3 | 1 |
| **Subtotal** | **14** | **21** | **21** |
| **G. Elective Courses** *(Choose any 2)* | | | |
| Engine Friction and Lubrication, | 2 | 0 | 2 |
| Solar Energy and Wind Utilization | 2 | 0 | 2 |
| Design of Thermal Systems, | 2 | 0 | 2 |
| Finite Element Method (FEM) and Computational Fluid Dynamics (CFD) | 2 | 0 | 2 |
| Introduction to Robotics and Automation, | 2 | 0 | 2 |
| **Subtotal** | **4** | **0** | **4** |
| **TOTAL TECHNICAL COURSES** | **108** | **93** | **146** |
| | | | |
| **II. NON-TECHNICAL COURSES** | | | |
| **A. General Education Courses** | | | |
| Understanding the Self | 3 | 0 | 3 |
| Mathematics in the Modern World | 3 | 0 | 3 |
| The Contemporary World | 3 | 0 | 3 |
| Readings in Philippine History | 3 | 0 | 3 |
| Purposive Communication | 3 | 0 | 3 |
| Ethics | 3 | 0 | 3 |
| Art Appreciation | 3 | 0 | 3 |
| Science, Technology and Society | 3 | 0 | 3 |
| **Subtotal** | **24** | **0** | **24** |
| **B. General Education Electives and Mandated Courses** | | | |
| Linear Algebra for AI | 3 | 0 | 3 |
| Machine Learning and Neural Networks | 2 | 3 | 3 |
| The Entrepreneurial Mind | 3 | 0 | 3 |
| Life and Works of Rizal | 3 | 0 | 3 |
| **Subtotal** | **11** | **3** | **12** |

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| | | | | |
|---|---|---|---|---|
| **C. Physical Activities Toward Health and Fitness (PATHFit)** | | | | |
| Movement Competency Training | 2 | 0 | 2 | |
| Exercise-based Fitness Activities | 2 | 0 | 2 | |
| Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1 | 2 | 0 | 2 | |
| Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2 | 2 | 0 | 2 | |
| **Subtotal** | **8** | **0** | **8** | |
| **D. National Service Training Program** | | | | |
| National Service Training Program 1 | 3 | 0 | 3 | |
| National Service Training Program 2 | 3 | 0 | 3 | |
| **Subtotal** | **6** | **0** | **6** | |
| **TOTAL NON-TECHNICAL COURSES** | **49** | **3** | **50** | |
| **GRAND TOTAL** | **157** | **96** | **196** | |

**SUMMARY**

| Courses | Number of Units |
|---|---|
| **I. TECHNICAL COURSES** | |
| A. Mathematics | 17 |
| B. Natural/Physical Sciences | 11 |
| C. Basic Engineering Sciences | 24 |
| D. Allied Courses | 9 |
| E. Fundamental ME Courses | 60 |
| F. Professional Courses | 21 |
| G. Elective Courses | 4 |
| **II. NON-TECHNICAL COURSES** | |
| A. General Education Courses | 24 |
| B. General Education Electives and Mandate Courses | 12 |
| C. Physical Activities Toward Health and Fitness (PATHFit) | 8 |
| D. National Service Training Program | 6 |
| **GRAND TOTAL** | **196** |

**CURRICULUM STRUCTURE**

**BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING**

Reference CMOs: CMO No. 97 s. 2017, CMO No. 20, s. 2013,
CMO No. 39, s. 2021, and CMO 40, s. 2021

| FIRST YEAR | | | | | | | |
|---|---|---|---|---|---|---|---|
| FIRST SEMESTER | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| MATH 101 | Differential Calculus | 4 | 4 | 0 | | | TC-A |
| ENGG 101 | Introduction to | 1 | 0 | 3 | | | TC-C |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Engineering | | | | | | |
| CHEM 101 | Chemistry for Engineers | 4 | 3 | 3 | | | TC-B |
| GEd 101 | Understanding the Self | 3 | 3 | 0 | | | NTC-A |
| GEd 102 | Mathematics in the Modern World | 3 | 3 | 0 | | | NTC-A |
| ENGG 102 | Engineering Drawing | 1 | 0 | 3 | | | TC-C |
| GEd 106 | Purposive Communication | 3 | 3 | 0 | | | NTC-A |
| PATHFit 1 | Movement Competency Training | 2 | 2 | 0 | | | NTC-C |
| NSTP 111 | National Service Training Program 1 | 3 | 3 | 0 | | | NTC-D |
| | **TOTAL** | **24** | **21** | **9** | | | |

| **FIRST YEAR** | | | | | | | |
|---|---|---|---|---|---|---|---|
| **SECOND SEMESTER** | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| MATH 102 | Integral Calculus | 4 | 4 | 0 | MATH 101 | | TC-A |
| GEd 105 | Readings in Philippine History | 3 | 3 | 0 | | | NTC-A |
| PHYS 102 | Physics for Engineers 1 | 4 | 3 | 3 | MATH 101 | MATH 102 | TC-B |
| GEd 104 | The Contemporary World | 3 | 3 | 0 | | | NTC-A |
| GEd 108 | Art Appreciation | 3 | 3 | 0 | | | NTC-A |
| GEd 109 | Science, Technology and Society | 3 | 3 | 0 | | | NTC-A |
| CpE 101 | Computer Programming | 1 | 0 | 3 | | | TC-C |
| PATHFit 2 | Exercise-Based Fitness Activities | 2 | 2 | 0 | PATHFit 1 | | NTC-C |
| NSTP 121 | National Service Training Program 2 | 3 | 3 | 0 | NSTP 111 | | NTC-D |
| | **TOTAL** | **26** | **24** | **6** | | | |

| **FIRST YEAR** | | | | | | | |
|---|---|---|---|---|---|---|---|
| **MIDTERM** | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| PHYS 103 | Physics for Engineers 2 | 3 | 2 | 3 | PHYS 102 | | TC-B |
| GEd 103 | Life and Works of Rizal | 3 | 3 | 0 | | | NTC-B |
| GEd 107 | Ethics | 3 | 3 | 0 | | | NTC-A |
| | **TOTAL** | **9** | **8** | **3** | | | |

| **SECOND YEAR** | | | | | | | |
|---|---|---|---|---|---|---|---|
| **FIRST SEMESTER** | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| GEdE 101 | The Entrepreneurial Mind | 3 | 3 | 0 | | | NTC-B |
| MATH 103 | Differential Equations | 3 | 3 | 0 | MATH 102 | | TC-A |
| ENGG 103 | Computer-Aided Design | 1 | 0 | 3 | ENGG 102 | | TC-C |
| ENGG 107 | Statics of Rigid Bodies | 3 | 3 | 0 | PHYS 102, MATH 102 | | TC-C |
| ME 201 | Mechanical Engineering Orientation | 1 | 1 | 0 | ENGG 101 | | TC-E |
| ME 202 | Thermodynamics 1 | 3 | 3 | 0 | PHYS 103, MATH 102 | | TC-E |
| ME 203 | Workshop Theory and Practice | 1 | 0 | 3 | | | TC-E |
| EE 205 | Basic Electrical Engineering | 3 | 2 | 3 | PHYS 103, MATH 102 | | TC-D |
| PATHFit 3 | Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 1 | 2 | 2 | 0 | PATHFit 1, PATHFit 2 | | NTC-C |
| AI 101 | Linear Algebra for AI | 3 | 3 | 0 | | | NTC-B |
| | **TOTAL** | **23** | **20** | **9** | | | |

| **SECOND YEAR** | | | | | | | |
|---|---|---|---|---|---|---|---|
| **SECOND SEMESTER** | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| ENGG 108 | Dynamics of Rigid Bodies | 2 | 2 | 0 | ENGG 107 | | TC-C |
| ENGG 111 | Mechanics of Deformable Bodies | 4 | 4 | 0 | ENGG 107 | ENGG 108 | TC-C |
| ME 204 | Advanced Mathematics for ME | 3 | 3 | 0 | MATH 103 | | TC-E |
| ME 205 | Thermodynamics 2 | 3 | 3 | 0 | ME 202 | | TC-E |
| ME 206 | Fluid Mechanics for ME | 3 | 3 | 0 | ME 202, ENGG 107 | | TC-E |
| ME 207 | Computer Applications for ME | 1 | 0 | 3 | ENGG 103, CpE 101 | | TC-E |
| ME 208 | Machine Shop Theory and Practice | 2 | 0 | 6 | ME 203 | | TC-E |
| EE 206 | DC and AC Machinery | 3 | 2 | 3 | EE 205 | | TC-D |
| PATHFit 4 | Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities 2 | 2 | 2 | 0 | PATHFit 1, PATHFit 2 | | NTC-C |
| AI 102 | Probability and Statistics for AI | 3 | 3 | 0 | Math 102 | | TC-A |
| | **TOTAL** | **26** | **22** | **12** | | | |

| **THIRD YEAR** | | | | | | | |
|---|---|---|---|---|---|---|---|
| **FIRST SEMESTER** | | | | | | | |
| **Course** | **Course Title** | **Credit** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-** | **Category** |

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| Code | | Units | | | | requisite(s) | |
|---|---|---|---|---|---|---|---|
| ENGG 116 | Methods of Research | 3 | 3 | 0 | AI 102 | | TC-E |
| MATH 104 | Numerical Methods | 3 | 3 | 0 | ME 204 | | TC-A |
| ME 301 | Heat Transfer | 3 | 3 | 0 | ME 204, ME 205 | | TC-E |
| ME 302 | Combustion Engineering | 3 | 3 | 0 | ME 205 | | TC-E |
| ME 303 | Fluid Machineries | 3 | 3 | 0 | ME 206 | | TC-E |
| ME 304 | Machine Elements | 3 | 2 | 3 | ENGG 108 | | TC-E |
| ME 305 | Materials Science and Engineering for ME | 3 | 2 | 3 | ENGG 111, CHEM 101 | | TC-E |
| ME 306 | ME Laboratory 1 | 1 | 0 | 3 | ME 205 | | TC-E |
| ME 307 | ME Elective 1 | 2 | 2 | 0 | Third Year Standing | | TC-G |
| ECE 406 | Basic Electronics | 3 | 2 | 3 | EE 205 | | TC-D |
| | **TOTAL** | **27** | **23** | **12** | | | |

| THIRD YEAR | | | | | | | |
|---|---|---|---|---|---|---|---|
| SECOND SEMESTER | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| ENGG 104 | Engineering Economics | 3 | 3 | 0 | Third Year Standing | | TC-C |
| ENGG 114 | Basic Occupational Safety and Health | 3 | 3 | 0 | Third Year Standing | | TC-E |
| ME 308 | Refrigeration Systems | 3 | 3 | 0 | ME 301 | | TC-E |
| ME 309 | Machine Design 1 | 3 | 3 | 0 | ME 304, ME 305 | ME 311 | TC-F |
| ME 310 | ME Laboratory 2 | 2 | 0 | 6 | ME 301 ME 303, ME 306 | | TC-E |
| ME 311 | ME Laws, Ethics, Contracts, Codes and Standards | 2 | 2 | 0 | GEd 107 | | TC-F |
| ME 312 | Vibration Engineering | 2 | 2 | 0 | MATH 103 | | TC-E |
| ME 313 | ME Project Study 1 | 1 | 0 | 3 | ENGG 116 | ME 309 | TC-F |
| AI 103 | Machine Learning and Neural Networks | 3 | 2 | 3 | AI 101, AI 102, CpE 101 | | NTC-B |
| | **TOTAL** | **22** | **18** | **12** | | | |

| THIRD YEAR | | | | | | | |
|---|---|---|---|---|---|---|---|
| MIDTERM | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| ME 314 | Air Conditioning and Ventilation Systems | 3 | 2 | 3 | ME 308 | | TC-E |
| ME 315 | Machine Design 2 | 3 | 2 | 3 | ME 309 | | TC-F |
| ME 316 | ME Elective 2 | 2 | 2 | 0 | Third Year Standing | | TC-G |
| | **TOTAL** | **8** | **6** | **6** | | | |

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| FOURTH YEAR | | | | | | | |
|---|---|---|---|---|---|---|---|
| FIRST SEMESTER | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| ENGG 117 | OJT/Internship (320 hrs) | 4 | | | 320 | Fourth Year Standing | |
| ENGG 113 | Environmental Science and Engineering | 3 | 3 | 0 | CHEM 101 | | TC-C |
| ME 401 | Power Plant Design with Renewable Energy | 5 | 4 | 3 | ME 302, ME 303, ENGG 104 | | TC-F |
| | **TOTAL** | **12** | **7** | **3** | | | |

| FOURTH YEAR | | | | | | | |
|---|---|---|---|---|---|---|---|
| SECOND SEMESTER | | | | | | | |
| **Course Code** | **Course Title** | **Credit Units** | **Lec** | **Lab** | **Prerequisite(s)** | **Co-requisite(s)** | **Category** |
| ENGG 105 | Technopreneurship | 3 | 3 | 0 | GEdE 101 | | TC-C |
| ENGG 106 | Engineering Management | 2 | 2 | 0 | Fourth Year Standing | | TC-C |
| ME 402 | Control Systems Engineering | 3 | 2 | 3 | ECE 406 | | TC-E |
| ME 403 | Industrial Plant Engineering | 4 | 3 | 3 | ME 314, ME 315, ME 401 | ME 402, ME 404 | TC-F |
| ME 404 | Manufacturing and Industrial Processes with Plant Visits | 2 | 1 | 3 | ENGG 114 | | TC-E |
| ME 405 | ME Laboratory 3 | 2 | 0 | 6 | ME 310,ME 314, ME 401, | | TC-F |
| ME 406 | ME Practice with Comprehensive Examination | 2 | 0 | 6 | Graduating | | TC-E |
| ME 407 | ME Project Study 2 | 1 | 0 | 3 | ME 313, ME 315, ENGG 104 | | TC-F |
| | **TOTAL** | **19** | **11** | **24** | | | |

**TOTAL UNITS: 196**